

Ocular Tumors

Visual Outcomes of Parapapillary Uveal Melanomas Following Proton Beam Therapy



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Summary

Parapapillary melanoma patients have been treated conservatively with proton therapy for 25 years. Patient, tumor, initial visual acuity, treatment, and radiation-induced complications influence visual acuity. However, a personalized computed tomography-based approach using a wedge filter to limit irradiation of the papilla and optic nerve length or macula

Purpose: In parapapillary melanoma patients, radiation-induced optic complications are frequent and visual acuity is often compromised. We investigated dose-effect relationships for the optic nerve with respect to visual acuity after proton therapy.

Methods and Materials: Of 5205 patients treated between 1991 and 2014, those treated using computed tomography (CT)-based planning to 52 Gy (prescribed dose, not accounting for relative biologic effectiveness correction of 1.1) in 4 fractions, with minimal 6-month follow-up and documented initial and last visual acuity, were included. Deterioration of ≥ 0.3 logMAR between initial and last visual acuity results was reported.

Results: A total of 865 consecutive patients were included. Median follow-up was 69 months, mean age was 61.7 years, tumor abutted the papilla in 35.1% of patients, and tumor-to-fovea distance was ≤ 3 mm in 74.2% of patients. Five-year relapse-free survival rate was 92.7%. Visual acuity was $\geq 20/200$ in 72.6% of patients initially and 47.2% at last follow-up. A wedge filter was used in 47.8% of the patients, with a

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when possible, reached a 5-year relapse-free survival rate of 92.7% and preservation of visual acuity over 20/200 in 61.1% of 865 parapapillary patients treated with proton therapy.

positive impact on vision and no impact on relapse. Glaucoma, radiation-induced optic neuropathy, maculopathy were reported in 17.9%, 47.5%, and 33.6% of patients, respectively. On multivariate analysis, age, diabetes, thickness, initial visual acuity and percentage of macula receiving 26 Gy were predictive of visual acuity. Furthermore, patients irradiated to $\geq 80\%$ of their papilla had better visual acuity when limiting the 50% (30-Gy) and 20% (12-Gy) isodoses to ≤ 2 mm and 6 mm of optic nerve length, respectively.

Conclusions: A personalized proton therapy plan with optic nerve and macular sparing can be used efficiently with good oncological and functional results in parapapillary melanoma patients. © 2016 Elsevier Inc. All rights reserved.

Introduction

Uveal melanoma is the most frequently diagnosed primary ocular tumor. Several studies (randomized and large retrospective ones) have validated the use of conservative treatments against enucleation owing to similar survival rates and no increased risk of metastasis with either treatment (1). Parapapillary melanomas refer to tumors close to the optic disk, whereas juxtapapillary melanomas refer to tumors abutting the optic disk, and both can be difficult to treat with brachytherapy (2). Hypofractionated proton therapy is performed in approximately 12 centers worldwide (3) and yields eye conservation and 5-year tumor relapse-free rates of approximately 90% and 94% to 96%, respectively (4, 5). In parapapillary melanoma patients, full dose to the papilla is commonly unavoidable, and radiation-induced optic neuropathy (RION), maculopathy, neovascular glaucoma (NVG), retinal detachment, and intraocular hemorrhage are associated with visual deterioration. Considering that the optic nerve has a serial architecture, full-dose irradiation of its section proximal to the retina theoretically leads to vision loss. However, it has been suggested that the optic nerve has a mixed behavior and that more complex dose volume effects have been reported with external beam irradiation (6). Moreover, observations suggest that patients receiving the maximal dose to their whole papillary surface with proton therapy recover some vision (7, 8). We investigated the impact of personalized optic nerve sparing despite papillary irradiation on visual acuity (VA) in a series of consecutive parapapillary melanoma patients treated with computed tomography (CT)-based proton therapy planning.

Methods and Materials

Treatment planning

Eye proton therapy started in 1991, using the 65-MeV dedicated hospital-based cyclotron of Nice, France. Four tantalum fiducial markers (clips) were positioned on the sclera under anesthesia, using transillumination (9, 10). Clip-to-clip, tumor-to-clips, tumor-to-macula and papilla distances, thicknesses, and diameters were reported in

ophthalmology reports based on transillumination, fundus (or retinography/angiography), and ultrasonography. Proton therapy was performed 2 to 4 weeks after clip placement. Irradiation was delivered at 52 Gy in 4 days (prescribed dose did not account for relative biologic effectiveness of 1.1, or ≈ 57 Gy cobalt gray equivalent [GCE]). This high-dose hypofractionated regimen has been empirically defined to eradicate melanomas (“radioresistant” tumors) and is used worldwide. Eyeplan treatment planning system software (versions 1-3) was used. It models the eye as a homogeneous ellipsoid with clips at its surface and moveable in space by a system of Cartesian and polar coordinates. Since 1994, ocular CT scans with clips have been performed to correct eye diameters to the patient’s and position the optic nerve relative to clips. CT also allowed delineation of the nerve and lens and correction of the eye model’s macular fixation to the patient fixation during first simulation. Fixation was optimized for preservation of ocular structures, and 2.5-mm margins were set by adjusting the range and modulation width of the spreadout Bragg peak plateau’s proximal and distal edges. Beam modifiers were designed depending on the target range, on the hypothesis of a rectilinear trajectory of protons. Lateral conformation was achieved by a brass collimator adjusted to the tumor contours, with 2.5-mm lateral safety margins. The papilla and nerve were preserved whenever possible without compromising tumor control. When possible, the macula was also preserved, even when the optic nerve could not be spared. If not achieved by optimization of fixation, a wedge filter could be used with an ultrasound transmission gel used as “missing tissue” compensator, creating a flat surface at the beam entrance in order to avoid irradiation of the posterior eye (Fig. 1). The gel included the eyelid, creating a steadily thick anterior tissue, thus avoiding variations of eyelid thickness (mainly due to lid edema) and uncertainties in range calculations from one session to another. The wedge effect was thus well controlled. Dose levels are reported in Eyeplan as 90%, 50%, and 20% of the total dose. Proper tumor and ocular structure projection relative to the proton beam axis during irradiation was achieved by focusing the patient’s gaze on a light-emitting diode located in a plane perpendicular to the beam. Positioning of the tumor in the beam during dosimetry and treatment phases was controlled with the

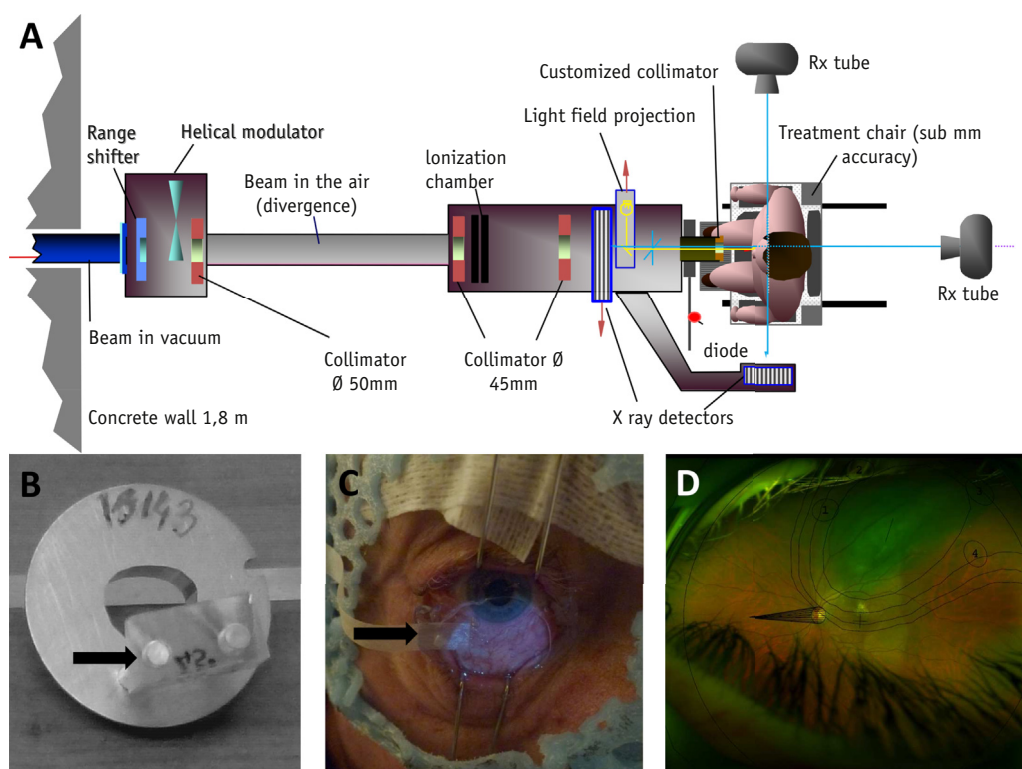


Fig. 1. (A) Treatment line including beam modifiers. (B) Brass collimator and wedge filter (arrow). (C) Gel compensator (arrow) and light projection of the proton beam (D) 90%, 50%, and 20% isodose lines around the melanoma and optic nerve (represented as a cone). The macula is represented as a cross and the tumor by using a wide-field retinogram.

clips using online orthogonal X-ray images. The eye position was monitored by video camera during the 10 second treatment, eye movements triggering immediate interruption of the beam.

Patient selection criteria

The overall prospectively constituted database consisted of 5205 eye tumor patients treated between 1991 and 2014. Patients treated for uveal melanomas at the proton therapy center from 1994 (ie, using CT-based planning) to 2013 and with minimal 6-month follow-up (ie, first onco-ophthalmologic assessment) were included in this retrospective study. Patients treated for conjunctival tumors, hemangiomas, and age-related macular degeneration and German uveal melanoma patients treated in 8 instead of 4 fractions, as well as patients referred from the other French center not allowing the use of a wedge filter, were excluded. Tumors were within 2 optic disk distances (ie, 3 mm from the papillary periphery). Owing to very small lateral penumbra and addition of 2.5 mm lateral margins, this distance represents a relevant threshold below which the optic disk cannot be spared completely and plan optimization is necessary to deliver as low as reasonably achievable dose to the papilla. Subgroups analyses were performed based on the papillary surface involved in the high dose, and nerve length was studied as a continuous variable or with 2- and

6-mm thresholds. Distance to the macula (center of fovea) and prognostic factors were investigated.

Outcome measures

Patients underwent visits with tumor assessment by an onco-ophthalmologist and liver ultrasonography every 6 months. Best corrected VA measurement using the Snellen scale and biomicroscopic examination of the fundus were performed at each control, with retinal photography and/or fluorescein angiography when necessary upon observations by the onco-ophthalmologist. Tumor thickness and diameter were measured at each visit with ocular echography. Interval consultations were performed by family ophthalmologists. Date and type of recurrences, secondary enucleation, and type of complications were collected.

Definition of visual acuity

Snellen scale data were converted into a logMAR scale by using the formula $[-\log\text{MAR VA}]$ with no light perception (3 logMAR) or light perception only (2.5 logMAR), hand motion (2.3 logMAR), or count fingers (1.9 logMAR). Initial and final VA values were available for analyses of VA. Differences were considered significant if there was more than ≥ 0.3 logMAR deterioration between initial and last VA.

Statistics

Follow-up time was defined as the time from the end of irradiation to the last visit. Time to last VA was defined as the time from completion of irradiation to the last VA available. A χ^2 tests and t test (or nonparametric test when needed) were used for comparison of rates and means. Survival curves were compared using log-rank tests. Univariate and multivariate analyses using the Cox proportional hazard model (backward Wald process) were performed using SPSS version 12 software. Confidence intervals (CI) of hazard ratio (HR) were estimated using normal approximation. Relapse-free survival (11, 12) and overall survival (OS) were calculated from tumor diagnosis to ocular relapse and death, respectively. Observations were censored at the last follow-up date. Statistical tests with *P* values smaller than 5% were considered significant.

Results

There were 4224 uveal melanoma patients (among 5205 ocular tumor patients). The selected 865 parapapillary melanoma patients were younger than the overall population. Diameter, thickness and stage were smaller and extrascleral extension, involvement of the ciliary body and iris root less frequent while retinal detachment was more frequent. Baseline VA was worse than in the overall population. Other initial characteristics were similar (data not shown). There was a center effect with respect to referral for posterior tumors ($P < .001$).

Parapapillary melanoma patient and disease characteristics

Mean age was 61.7 years old (range: 13.0-93.0), 6.6% and 17.2% of patients had diabetes and hypertension, respectively, and 52.2% were male.

Median maximal diameter and thickness were 14.40 mm (range: 4.1-24.0 mm) and 4.90 mm (range: 1.0-14.70 mm), respectively. Diameter was ≥ 15 mm, and thickness was ≥ 6 mm in 45.1% and 36.6% of patients, respectively. Median distances to the papilla and macula were 0.90 (range: 0.00-3.00) and 0.90 mm (range: 0.00-8.00), respectively. Of those, the tumor abutted the papilla in 35.1% of cases and tumor-to-fovea distance was ≤ 3 mm in 74.2% of patients. Peritumoral, inferior, extensive retinal detachment and intravitreal hemorrhage were reported in 35.0%, 15.7%, 2.1%, and 7.4% of the patients, respectively. Some degree of cataract was present in 22.6% of patients, and 9.8% were pseudophakic. Before proton therapy, 1% and 72.6% of patients had no light perception and $\geq 20/200$ VA, respectively.

Median proximal and distal margins were 2.5 mm (2.0, 0.0-3.5 mm) with a 26.5-mm range (26.45, 2.1-31.2 mm) and 16.80-mm modulation (0.0-28.9 mm). A wedge filter was used in 48.6% of the patients. The papillary surface

receiving 90% of the dose (47 Gy) was $<20\%$ [20 to $<50\%$], [50 to $<80\%$], [80 to $<100\%$] and 100% in 35.9%, 8.7%, 9.5%, 3.6%, and 42.3% of the patients, respectively. Mean lateral margins and field size were 2.5 mm (range: 0.0-3.5 mm) and 182.0 mm² (16.6-482.0 mm), respectively.

On average, 54.9%, 66.2%, and 77.2% of the papillary surface received 90% (47 Gy), 50% (26 Gy), and 20% of the total dose (10 Gy), respectively. Mean optic nerve length irradiated within the 90% isodose was 1.45 mm. The 50% isodose covered 1.96 mm of the proximal nerve length (26 Gy) and the 20% isodose 2.6 mm (10 Gy). Dose to the macula and dose volume effects on VA are reported in Table 1.

Outcomes

Mean follow-up was 69.0 months (median: 53.3 months; range: 6.0-240.0 months) including 21.3%, 34.0%, 27.2%, and 17.5% of patients with 6 months to 2 years, 2 to 5 year, 5 to 10 year, and over 10-year follow-up, respectively. Crude local recurrence rate was 10.5%. Crude metastatic rate was 18.2%. Crude overall death rate was 20.9%. Median and 2-, 5-, 10-, and 15-year survival rates were 228.9 months and 94.5%, 82.4%, 69.7%, and 57.7%, respectively. Median and 2-, 5-, 10-, and 15-year metastasis-free survival rates were 236.9 months and 98.5%, 95.6%, 70%, and 55.4%, respectively. Mean and 2-, 5-, and 10-year relapse-free survival rates were 224.5 months (median not reached) and 96.6%, 92.7%, and 88.8%, respectively (Fig. 2). Crude enucleation rate of 12% and was due to relapse. The use of a wedge filter to spare the vision was not associated with worse relapse rates ($P = .97$). Other prognostic factors for survival outcomes are not the focus of this article and were published elsewhere (11, 12).

During follow-up, 11.5% of patients experienced intra-vitreous hemorrhage or hyphema, resulting in poorly assessable fundus in 8.7% with 4.2% requiring vitrectomy. Peritumoral, inferior and extensive retinal detachment was reported in 12.5%, 1.1%, and 1.6% of the patients, respectively. Some degree of cataract was present in 28.7% (+6.1% compared to baseline) of the patients. Some degree of inflammation was present in 10.5% of the patients. Dry eye syndrome was reported in 9.4% of patients, with focal corneal ulceration in 0.4%. NVG, RION, and maculopathy were reported in 17.9%, 47.5%, and 33.6% of patients at last follow-up, respectively.

With respect to initial VA, 80.5% of the patients could at least experience light perception, and 61.1% had $\leq +1.0$ logMAR ($\geq 20/200$) before proton therapy. At last follow-up assessment, 80.8% of patients had some light perception and 47.2% of patients had $\leq +1.0$ logMAR ($\geq 20/200$). Of 7 patients with no light perception before proton therapy, 3 had recovered some vision at last follow-up assessment, although it remained $<20/200$. Mean and median VA changes were 0.89 and 0.70 logMAR, respectively (± 1.00)

Table 1 Initial prognostic factors for vision loss ≥ 0.3 logMAR using the log-rank test and Kaplan-Meier curves

Factor	Categories	Univariate hazard ratio (95% CI, <i>P</i> value)	Multivariate hazard ratio (95% CI, <i>P</i> value) for significant values in general model
Patient characteristics			
Age	<Median*/≥median	1.018 (1.013-1.022, <i>P</i> <.0001)	1.013 (1.005-1.021, <i>P</i> =.002)
Sex	Male*/female	0.921 (0.777-1.092, <i>P</i> =.343)	
Diabetes	Yes/no*	1.817 (1.2156-2.717, <i>P</i> =.004)	
Hypertension	Yes/no*	1.602 (1.221-2.101, <i>P</i> =.001)	
Baseline tumor characteristics			
Diameter	<15 mm*/≥15 mm	1.075 (1.053-1.098, <i>P</i> <.001)	1.093 (1.042-1.147, <i>P</i> <.001)
Thickness	<6 mm*/≥6 mm	1.136 (1.114-1.158, <i>P</i> <.001)	
Stage	T1/2/3/4	<i>P</i> <.0001	
		1.298 (0.991-1.699, <i>P</i> =.058)	
		1.744 (1.330-2.288, <i>P</i> <.0001)	0.651 (0.543-0.780, <i>P</i> <.0001)
		2.498 (1.883-3.313, <i>P</i> <.001)	
Retinal detachment	Absent	<i>P</i> <.0001	
	Peritumoral	0.912 (0.728-1.142, <i>P</i> =.421)	
	Inferior	1.415 (1.071-1.868, <i>P</i> =.014)	0.651 (0.543-0.780, <i>P</i> <.0001)
	Extended	3.570 (1.136-11.222, <i>P</i> =.029)	
Tumor ≤ 3 mm of fovea	Continuous	0.992 (0.950-1.036, <i>P</i> =.715)	
	Yes/no*	0.755 (0.674-0.846, <i>P</i> <.0001)	
Initial visual acuity	Continuous	0.998 (0.911-1.094, <i>P</i> =.972)	0.651 (0.543-0.780, <i>P</i> <.0001)
	≥ 20/200	0.841 (0.718-0.986, <i>P</i> =.033)	
	≥ light perception	0.049 (0.001-2.115, <i>P</i> =.116)	
Treatment parameters			
Wedge	Yes*/no	1.234 (1.040-1.464, <i>P</i> =.016)	1.006 (1.003-1.009, <i>P</i> <.0001)
Papillary surface in isodose	90%	1.004 (1.002-1.006, <i>P</i> <.001)	
	50%	1.005 (1.003-1.007, <i>P</i> <.001)	
	20%*	1.005 (1.003-1.014, <i>P</i> =.004)	
Papillary surface in isodose	< 20%*	<i>P</i> <.001	1.006 (1.003-1.009, <i>P</i> <.0001)
	20 to <50%	1.639 (1.428-1.881, <i>P</i> <.0001)	
	50 to <80%	1.637 (1.233-2.175, <i>P</i> =.001)	
	80 to 100%	1.290 (0.961-1.732, <i>P</i> =.089)	
ONL in 90% isodose	< 2 mm*/≥ 2 mm	1.120 (1.067-1.176, <i>P</i> <.0001)	1.006 (1.003-1.009, <i>P</i> <.0001)
ONL in 50% isodose	<2 mm*/≥2 mm	1.113 (1.066-1.162, <i>P</i> <.0001)	
ONL in 20% isodose	<2 mm*/≥2 mm	1.175 (1.075-1.284, <i>P</i> <.0001)	
ONL in 20% isodose*	<6 mm*/≥6 mm	1.120 (1.067-1.176, <i>P</i> <.0001)	
Dose to the macula	90% (47 Gy)	1.003 (1.001-1.005, <i>P</i> =.002)	1.006 (1.003-1.009, <i>P</i> <.0001)
	50% (26 Gy)	1.004 (1.002-1.006, <i>P</i> <.001)	
	20% (10 Gy)	1.003 (0.998-1.007, <i>P</i> =.279)	
Outcomes			
Cataract	Yes/no or implant*	0.938 (0.833-1.057, <i>P</i> =.293)	1.006 (1.003-1.009, <i>P</i> <.0001)
RION	Yes/no*	1.504 (1.053-2.148, <i>P</i> =.025)	
Retinopathy	Yes/no*	0.805 (0.633-1.024, <i>P</i> =.077)	
Maculopathy	Yes/no*	1.266 (0.858-1.866, <i>P</i> =.235)	
Vasculopathy	Yes/no*	0.928 (0.750-1.150, <i>P</i> =.496)	1.006 (1.003-1.009, <i>P</i> <.0001)
Retinal detachment	Absent*	<i>P</i> <.0001	
	Peritumoral	1.255 (0.801-1.967, <i>P</i> =.322)	
	Inferior	40.155 (11.188-144.119, <i>P</i> <.0001)	
	Subtotal	70.893 (14.815-339.241, <i>P</i> <.001)	1.006 (1.003-1.009, <i>P</i> <.0001)
		42.109 (5.168-343080, <i>P</i> <.001)	
Neovascular glaucoma	Yes/no*	1.289 (1.090-1.523, <i>P</i> =.003)	
Intraocular hemorrhage	Yes/no*	1.165 (0.667-2.035, <i>P</i> =.592)	

Abbreviations: ONL = optic nerve length; RION = radiation-induced optic neuropathy.

* Indicates reference category.

(Fig. 3). Comparing final to baseline VA, 61.7% of patients had lost >0.3 logMAR of VA, and 38.3% had steady or better VA. Of those patients losing VA, 44.6%, 66.6%,

68.1%, and 64.2% had lost more than >0.3 logMAR of VA within the first 2 years, between 2 and 5 years, 5 and 10 years, and >10 years (*P*<.0001), respectively (Fig. 4).

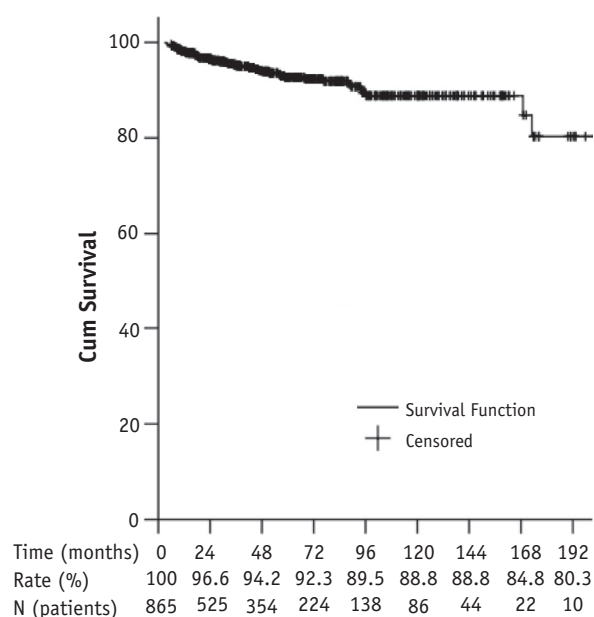


Fig. 2. Relapse-free survival is shown for 24, 48, 72, 96, 120, 144, 168, and 192 months; mean relapse-free $\pm$ standard deviation (Std. Dev.) values were 96.6 ± 0.72 , 94.2 ± 1.0 , 92.3 ± 1.2 , 89.5 ± 1.7 , 88.8 ± 1.9 , 88.8 ± 1.9 , and 80.3 ± 6.0 , respectively. Survival curves are truncated after fewer than 10 patients remained for analyses.

VA based on longest follow-up time suggests that most VA changes occurred between 2 and 5 years and that some patients might have recovered some vision after 5 years. Of patients with VA $>+1.0$ logMAR ($<20/200$), 17.5% had better VA at last follow-up.

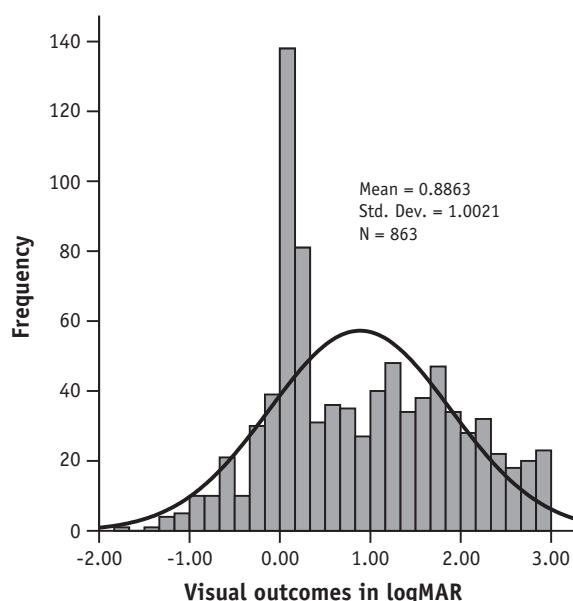


Fig. 3. Differential visual outcomes (in logMAR) at last follow-up are compared to baseline visual acuity.

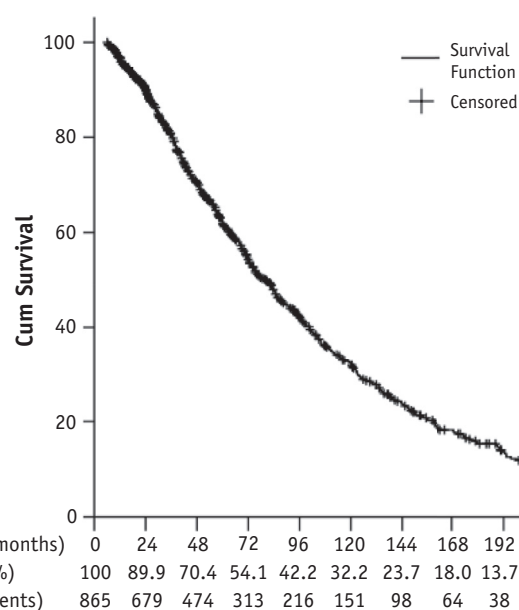


Fig. 4. Time to visual deterioration (ie, patients experiencing ≥ 0.3 logMAR difference in visual acuity) is shown. For the time points 24, 48, 72, 96, 120, 144, 168, and 192 months), mean $\pm$ Ste. Dev. (SD, standard deviation) visual deterioration-free and standard deviation were 89.9 ± 1.1 , 70.4 ± 1.7 , 54.1 ± 1.9 , 42.2 ± 2.0 , 32.2 ± 2.0 , 23.7 ± 2.9 , 18.0 ± 1.8 , and 13.7 ± 1.7 . Survival curves are truncated after less than 10 patients remained for analyses based on Fig. 2.

Prognostic factors for visual outcomes

The larger the papillary surface irradiated, the worse the VA deterioration (1.25 ± 0.06 logMAR for 20% of the papillary surface to 2.00 ± 0.05 logMAR for full papillary irradiation, $P < .0001$). However, of 364 patients undergoing full papillary irradiation, 252 (69%) had non-null nominal VA at last follow-up. Kaplan-Meier estimates using VA deterioration of ≥ 0.3 logMAR as an event, showed that patients with 100% or $\geq 80\%$ of papillary surface irradiated only trended toward better visual outcomes with limitation of optic nerve length irradiation ≤ 2 mm versus >2 mm ($P = .29$ and $.30$, respectively). However, VA was better when the optic nerve length covered by the 50% isodose (26 Gy) was ≤ 2 mm ($P = .0079$) when patients had $\geq 80\%$ of papillary surface irradiated. There was a trend for better VA in patients with $\geq 80\%$ of papillary surface irradiated when the optic nerve length covered by the 20% isodose (10 Gy) was limited to ≤ 6 mm ($P = .054$). Other categories were either non significant or had too few patient numbers in each.

Univariate analysis

In patients experiencing ≥ 0.3 logMAR differences in visual outcomes, prognostic factors investigated for visual deterioration are shown in Table 1.

Multivariate analysis

Age, diabetes, hypertension, diameter, thickness, distance to papilla and fovea, initial retinal detachment, initial VA, wedge filter, and percentage of macula covered by the different isodoses were entered in the model. Age, diabetes, thickness, initial VA and percent macula receiving 26 Gy remained in the model for visual deterioration ($P < .0001$) (Table 1).

Entering complications in the model, initial VA ($P = .005$), percentage of macula irradiated to 26 Gy ($P = .047$), NVG ($P = .001$), and RION ($P = .018$) were significantly associated with VA deterioration ($P < .0001$). Limiting analyses to patients with $\geq 80\%$ of their papillary surface irradiated to full dose, only optic nerve length ($P = .042$) receiving 10 Gy (20% isodose line) or 26 Gy (50% isodose line) remained in the model. In contrast, in patients with $< 80\%$ of papillary surface irradiated to full dose, age ($P = .021$), diabetes ($P = .003$), tumor thickness ($P < .0001$), percent of macula ($P < .0001$) receiving full dose were in the model ($P < .0001$).

To further focus on effects of optic nerve damage, similar analyses were performed in patients experiencing vision loss without residual light perception. Age ($P < .0001$), diameter ($P = .006$), and percentage of macular area covered by the 50% isodose ($P = .008$) remained significant on univariate analyses. Only NVG remained significant on multivariate analysis ($P = .01$).

Discussion

Consistent with previous observations, this 865-patient series suggests that effective visual preservation can be obtained in parapatapillary melanomas treated with hypofractionated proton therapy, with good local control (7, 9, 13, 14). In patients with baseline VA of $\geq 20/200$ in the treated eye, Lane et al (13) showed that some vision was retained in 50% and 20% of patients at 2 and 5 years after irradiation and that 50% could count fingers (13). Similar to the study by Meyer et al (7), we chose 2 disk distances and 2.5-mm lateral margins around the tumor borders. Within that distance, our strategy to preserve the nerve could be fully investigated, as no visual preservation could be achieved without optimizing the fixation and/or using a wedge filter for parapatapillary/juxtapapillary melanomas. The addition of a wedge filter is controversial because of the putative risk of local failure (eye set up and physics uncertainties). In contrast, the risk of direct nerve involvement has not been considered an issue as it is extremely low. We found no increased risk of failure in this series. VA after proton therapy was dependent on age, diabetes, initial VA, and optic nerve length. Percentage of macula was more marginally associated with VA. Of patients undergoing high-dose irradiation to full papillary surface, 69% retained non-null VA with a median 69-month follow-up. This compared with 64% with a 37-month follow-up (7). Our

data suggested that patients with full dose to 100% or $\geq 80\%$ of papillary surface, could be spared some vision by using optimization of optic nerve length irradiation. VA appeared better preserved when the 50% isodose could be limited to ≤ 2 mm of nerve length. This is consistent with published data in a series not restricted to parapatapillary melanomas (7). Of note, ways to reduce optic nerve length irradiation may include optimization of patient gaze, addition of a wedge filter and gel compensator, CT-based planning accounting for the true position of the optic nerve compared to a native eye model, and reduction of distal margins and/or of lateral margins. We previously showed that margin reduction ≤ 1 mm increases the risk of local relapse (12). It was beyond the scope of this article to show that we may obtain more accurate dose-volume histograms of the optic nerve. VA preservation may also be achieved by reducing the dose per fraction or decreasing the total dose. A modified hypofractionation trial randomized 70 participants to receive 50 Gy in 5 fractions (15). There was a trend toward lower papillopathy rates and less visual field loss with the lower dose.

We aimed to determine the utility of nerve length sparing despite high dose to the papilla as it might complicate our treatment planning process. One limitation of our study is that preservation strategy was systematic. As a consequence, we had no control group to rigorously assess the safety of our strategy. However, the high local control rate found here, including one-third of the tumors abutting the optic nerve, compared to other data for parapatapillary melanomas (7, 8, 13, 14), suggested that there was no excess risk with our strategy. However, there are some situations where changing the fixation (limited by the extent of the patient's gaze and nasal tumor location, for example) or adding a wedge filter, or both, will not help spare the optic nerve.

Despite the dominant serial architecture of the optic nerve, the impact of reducing the moderate and low dose levels over a certain nerve length (and volume) despite full dose irradiation to most of the papillary surface also suggests a parallel architecture (6). It is believed that the 2 mm of optic nerve proximal to the retina are an area of relative frailty owing to the absence of myelinated fibers. The 2-mm distance also corresponds to a vascular node at the junction between the retrograde central retinal artery and antero-grade flux of the short posterior ciliary arteries.

The correlation between RION and VA was significant but weaker than in other series. Reasons for this include underreporting or nonstandardized assessment of RION. The OPTIC team actively promotes the standardized reporting of radiation-induced complications and participation of radiation oncologists in the follow-up procedures (J. Hrbacek, et al, unpublished data 2016). It may also be hypothesized that our preservation strategy was efficient in limiting RION rates. Intriguingly, maculopathy was not associated with visual deterioration in this series while percent of macular area irradiated to moderate dose was marginally significant (especially if we considered more

stringent *P* significance threshold to account for multiple analyses). This is in contrast with the literature (16, 17). A fundus view is not sufficient to diagnose ischemic maculopathy and it is possible that we reported edematous maculopathy but underreported ischemic maculopathy by not using systematic angiography during follow-up. It is also possible that our patient selection (excluding melanomas ≥ 3 mm from papilla) and our strategy to spare the macula could partially explain that observation.

In contrast, NVG was highly associated with visual deterioration. The NVG risk correlates with proximity to the papilla (18). Different strategies are being implemented to improve the management of such complications.

Another limitation of our study is that interval VA assessments were not available. We deliberately included patients with minimal follow-up of 6 months. Hitherto, we estimated the kinetics of visual outcomes after proton therapy. We could assess that most patients experienced a decline of over 0.3 logMAR VA between 2 and 5 years. This will have to be confirmed with proper VA kinetics assessment using interval VA.

Our population showed higher tumor diameters than commonly found for posterior tumors in the literature (19). This diameter was defined based on ultrasonography, fundus measurements, and clips-to-tumor distances. In cases of inconsistency, a security strategy was adopted by choosing the worst case scenario, while respecting the tumor to papillary border distance scrupulously.

Conclusions

VA is a complex endpoint that depends on factors other than just the dose to the papilla and optic nerve length. Similarly, the optic nerve has a complex behavior as some patients recover some vision 5 years after the onset of papillopathy (8). We showed that other initial parameters, either related to the patient or disease, and complications highly influenced VA. Optic nerve length sparing appeared relevant. More systematic use of angiography during follow-up would probably help to standardize reporting of maculopathy and might show a stronger impact on visual outcomes, as we found the impact of the percent of the macula that is irradiated but not maculopathy. Additionally, other radiation-induced complications, such as NVG, highly influence VA. Management of those complications has largely improved in the latest years and is promising for the visual outcomes of patients with parapapillary melanomas treated with proton therapy.

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